



GIRIJANANDA CHOWDHURY UNIVERSITY, GUWAHATI
DEPARTMENT OF CIVIL ENGINEERING

Formative Assessment - I
Geotechnical Engineering (B. Tech 4th Semester)
Course: BCE23403T

Academic Year: 2025-2026
Total Marks: 20

27th March 2026
Duration: 1 hour

Questions	Marks	CO	Cognitive Level
1 a) Define C_c and D_{10} . b) What do MH and SC mean? c) Define thixotropy of clays. d) Draw the structural unit of Kaolinite and Montmorillonite. e) Define quick sand condition.	$1 \times 5 = 5$	1	Understand
2 An undisturbed sample of soil has a volume of 110 cm^3 and mass of 150 g. On oven drying for 24 hours, the mass is reduced to 110 g. If the specific gravity of grains is 2.68, determine the water content, void ratio and degree of saturation of the soil.	5	2	Apply
3 Draw the total stress, neutral stress and effective stress distribution of a granular soil deposit 8 m deep. The ground water table is at 3m below the ground surface. The deposit has a zone of capillary rise of 1.5 m with a saturation of 60%. Assume, unit weight of water as 10 kN/m^3 and $G=2.65$	$e = 0.6$ 5 Assume, $e = 0.6$	2	Apply
4 A horizontal stratified soil deposit consists of 3 uniform layers of thickness 6, 4 and 12 m respectively. The permeability of these layers are $8 \times 10^{-4} \text{ cm/s}$, $52 \times 10^{-4} \text{ cm/s}$ and $6 \times 10^{-4} \text{ cm/s}$, find the effective average permeability of the deposit in the horizontal and vertical direction.	5	2	Apply